Mahmoud Adas

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About Me

C++ SWE with 4 years building desktop apps in Qt/QML, most recently leading the general-CAD team at Atomica.ai. I also wrote some of the internal tools. On my own time I write a NES emulator and a flight-sim clone.

Education

Cairo University, Faculty of Engineering | *Egypt* **Sep 2015 - Jul 2021**
B.S. in Computer Engineering

Tools

- **C++**, **C**, Go, Python, Java, Rust
- **Qt/QML**, React, Spring, Android
- SQLite, Docker, Linux

Experience

C++ Software Engineer **Jun 2021 - Mar 2022, Jun 2023 - Now**
Atomica.ai (360Imaging) - Giza, Egypt

- Led the general-CAD team: set priorities, scheduled features, and shipped releases
- Maintained the Qt/QML frontend of the implant-planning and teeth-segmentation apps, integrating features from other teams and fixing bugs
- Built an ImGui/SQLite Library Editor so product managers could edit dental catalog data (implants, abutments, materials) and push it to AWS - catalog updates no longer needed an app release
- Wrote ldb2cpp, a SQLite-to-C++ generator for type-safe bindings; other internal projects adopted it
- Contributed to mn and ndcore, the shared C++ libraries behind every company product

Software Engineering Intern · Cert **Jul - Sep 2020**
Solectra - El Gouna, Egypt

- Maintained an **embedded system** that collects and analyses data from electric vehicles to report trips, total revenue, and energy
- Fixed bugs in the embedded telemetry system, improved trip/revenue prediction accuracy, and added tests

Projects

360Launcher **Jan - Mar 2022**
Qt/QML/C++ app to install/update/launch main products and integrate with them

DPSX-Sync **Jun - Aug 2021**

- Qt/QML/C++ experimental Dropbox-like app to add sync and collaboration features
- Uses libgit and Gitea server for versioning and synchronization

C4IAN *(Graduation Project)* · Repo **Feb - Jul 2021**

- Tactical Mobile ad-hoc network. Supports video/audio streaming and messaging
- ZHLS+ZID unicast, ODMRP multicast, and geo-based broadcast
- **Golang** router that integrates with the Linux network stack

OpenYSF · Repo **Nov 2021 - Present**
A ysflight flight simulator clone, backward compatible with existing ysflight game assets.

Apex · Backend · Frontend · Android · Testing **Jan - Mar 2019**

- Reddit clone, for web and android. Took role of **Devops** in team of 18
- Used Travis-CI for CI/CD, **GoogleCloud**, **Docker**, DockerCompose, and Datadog for monitoring

NesEmu · Repo **2019 - Present**

- NES emulator in C++: 6502 CPU (full instruction set), PPU, iNES ROM loading
- ImGui debugger with memory inspection, live disassembly, and tile/palette viewers. Built on SFML.

Koos · Repo · Rust Rewrite **Jun 2019**

- Decentralized social media app, built on Fluence. AssemblyScript/Rust for server, React for frontend

- Won a Gitcoin bounty for the best app that uses Fluence's technology

Match Reservation System · **Repo** **Jan - Feb 2021**
Web application for Egyptian Premier League for seats reservation. Used **MongoDB, Nodejs** and React

Mini-Search-Engine · **Repo** **Feb - Mar 2020**
Text and image search with trends detection. Used Java Spring and **React**

Cartoonization · **Repo** **Oct - Dec 2019**

- Snapchat Clone written in Python draws stickers on live recordings of faces
- Implemented GUI, rendering stickers, and HOG algorithm
- Used Viola-jones/Hog to detect faces, and multiple **image processing** techniques to filter the input

RetroSoccer · **Repo** **Jan - Mar 2019**
Multiplayer soccer game entirely in x86 IA32 MASM assembly with chat (serial communication)

Simple PDP--11 · **Repo** **Oct - Dec 2019**

- Design and simulation of a CPU architecture similar to *PDP--11* in VHDL and Vunit for test benches
- Designed and implemented automated tests for the CPU

TheMistake · **Repo** **Jan - Mar 2019**

- Maze solver car using flood fill algorithm on TIVA-C μc/Arduino-Uno with ultrasonic and IR sensors
- Implemented maze-solving and tested it